

Week 3 Submission: Data & Model Brief

Team Pulse

Section 0: Product Description

Product Name: PricePilot: *Autonomous AI pricing and offer optimization for revenue-focused SMEs*

Customer Painpoint: Small and medium enterprises (SMEs) frequently misprice products and services because they rely on static price lists, intuition, or manual competitor checks. As a result, prices lag in real-time market demand.

Product Description: PricePilot helps small and medium enterprises optimize pricing using machine learning, reducing revenue loss and manual effort caused by static pricing and slow reactions to change in demand.

Section 1: Data Requirements

1.1 Required Data

To function effectively, the platform requires:

- **Internal Business Data:** Transaction history (price, quantity, timestamp), product/service metadata (SKU, category, cost), inventory levels, and refund/cancellation records.
- **Customer Behavior Data:** Page views, click-through rates, cart abandonment/booking drop-offs, conversion rates by price point, and customer loyalty indicators (returning customers).
- **Competitive & External Data:** Competitor pricing signals, seasonality, inflation, and local event-based demand (holidays, weather).

1.2 Data Formats

- **Structured:** Transactions, inventory, and pricing tables.
- **Time-series:** Sales volume and demand fluctuations.
- **Semi-structured:** Competitor price feeds and logs.
- **Aggregated signals:** Seasonality indices and demand trends.

1.3 Data Volume Estimates

The system uses a chronological split to prevent data leakage and simulate real deployment.

- **Minimum Viable Product (MVP):** 6–12 months of transaction history (~1,000–2,000 transactions per business). Training is augmented using ~10,000–50,000 transactions of pooled, anonymized data per vertical. **Split:** 80% train / 10% validation / 10% test.
- **Production Scale:** Millions of transactions across multiple SMEs to enable fine-grained pricing by SKU and customer segment. Split: 85–90% train / 5–7.5% validation / 5–7.5% test

Section 2: Data Sources

2.1 Sources and Reliability

Data Type	Source	Quality & Reliability
Sales & Inventory	POS systems (Square, Toast), e-commerce (Shopify, WooCommerce)	High – System-generated and consistent
Customer Behavior	Analytics tools (Google Analytics, platform-native)	Medium-High – Depends on instrumentation
Competitor Pricing	Public listings, price aggregators, scraping APIs	Medium – Incomplete and noisy
External Signals	Public calendars, weather APIs, seasonality datasets	High – Stable and well-documented

2.2 Potential Sources of Bias

- **Survivorship Bias:** Historical data reflects past pricing decisions, not necessarily optimal ones.
- **Demand Distortion:** Promotions or discounts can skew elasticity estimates.
- **Seasonality Bias:** Short-term spikes (holidays) may be misinterpreted as long-term elasticity.
- **Mix & Geographic Bias:** Urban-heavy datasets or specific pack sizes may not generalize across all markets.

Mitigation: Can be addressed through normalization, confidence weighting, minimum price guardrails, and ongoing evaluation.

Section 3: Model Approach

3.1 Model Types

PricePilot uses a **hybrid modeling approach**:

- **Regression Models:** To estimate price elasticity under various conditions.
- **Time-series Forecasting:** To predict short-term demand trends and temporal patterns.
- **Optimization Models:** To recommend prices while respecting business constraints like minimum margins and inventory targets.
- **Reinforcement Learning (Future):** To optimize long-term revenue and demand stability.
- **Generative AI:** Used as a supporting layer to translate recommendations into natural-language explanations for users to increase trust.

3.2 Build vs. Buy Decision

- **Build (In-house):** Core pricing and optimization models are built in-house to capture domain-specific logic, constraints, and vertical nuances.
- **Buy/Off-the-shelf:** Leveraging established forecasting libraries, cloud ML platforms (for scaling/deployment), and pre-trained Large Language Models for the generative explanation layer. This balances customization with cost efficiency.

3.3 Key Technical Risks & Unknowns

- **Cold Start:** Limited data for new businesses is mitigated by using pooled anonymized data and conservative pricing.
- **Over-optimization:** Frequent price changes are managed via rate limits and human overrides.
- **Race-to-the-bottom:** Mitigated by minimum margin and value-based constraints.
- **Model Drift:** Changing demand patterns are addressed through continuous monitoring and retraining on sliding historical windows.
- **Trust:** Increased through transparent evaluation using **business-oriented metrics** (revenue lift, margin impact) rather than just model accuracy.